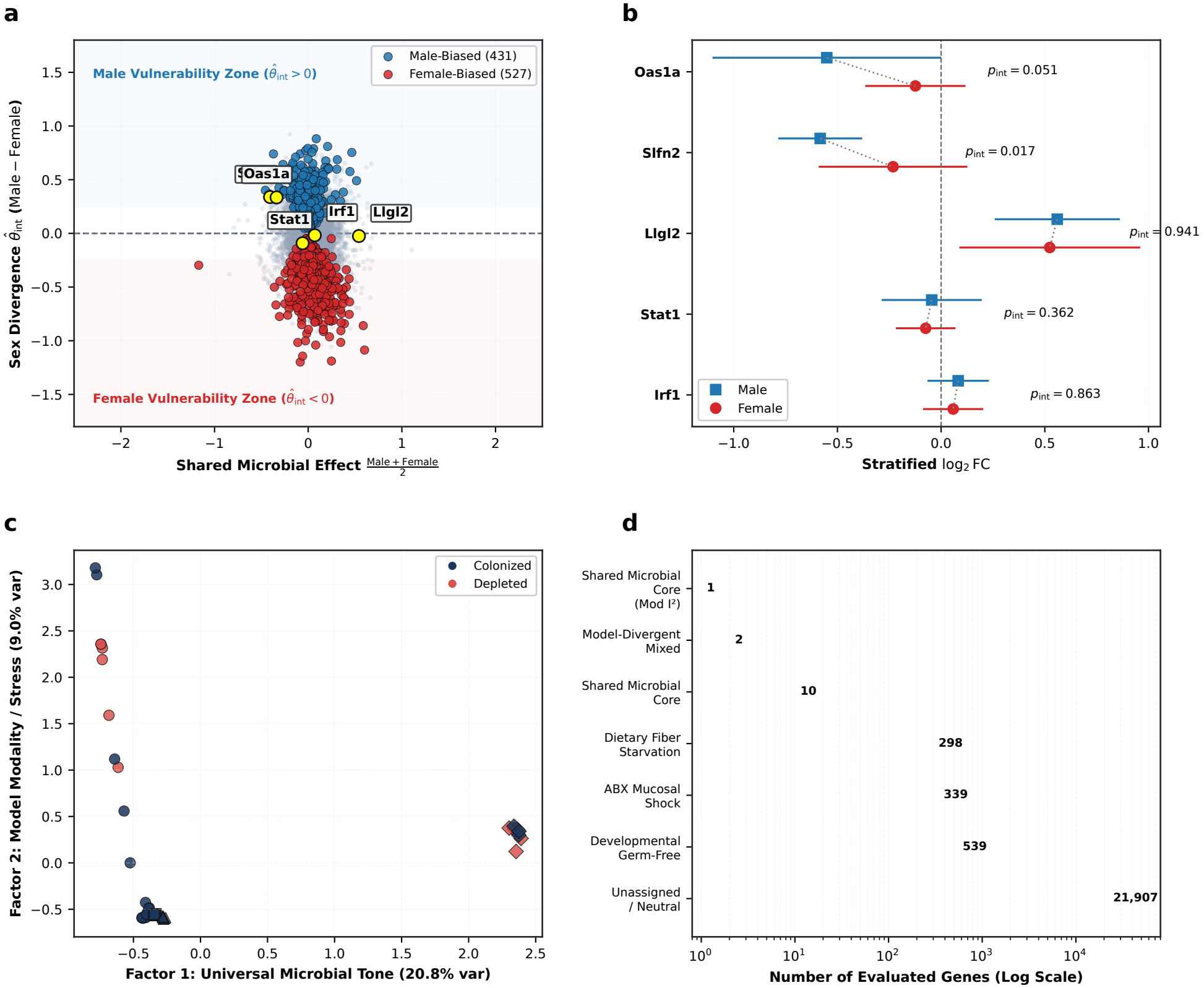

a
Male Vulnerability Zone ($\hat{\theta}_{int} > 0$)
Female Vulnerability Zone ($\hat{\theta}_{int} < 0$)
Male-Biased (431)
Female-Biased (527)
Oas1a
Stat1
Irf1
Llgl2
Sex Divergence $\hat{\theta}_{int}$ (Male − Female)
Shared Microbial Effect $\frac{Male + Female}{2}$
b
Oas1a
Slfn2
Llgl2
Stat1
Irf1
$p_{int} = 0.051$
$p_{int} = 0.017$
$p_{int} = 0.941$
$p_{int} = 0.362$
$p_{int} = 0.863$
Male
Female
Stratified $\log_2$ FC
c
Colonized
Depleted
Factor 2: Model Modality / Stress (9.0% var)
Factor 1: Universal Microbial Tone (20.8% var)
d
Shared Microbial Core (Mod I²)
Model-Divergent Mixed
Shared Microbial Core
Dietary Fiber Starvation
ABX Mucosal Shock
Developmental Germ-Free
Unassigned / Neutral
1
2
10
298
339
539
21,907
Number of Evaluated Genes (Log Scale)